Business Intelligence Internship Assignment – Approach & Assumptions

This document outlines the methodology, data generation logic, analytical approach, and assumptions used to complete the Business Intelligence assignment related to car insurance policy sales and claims analysis.

1. Business Context

The project simulates an insurance company's portfolio of car insurance policies sold during 2024. The objective is to generate realistic datasets for policy sales and claims activity in order to analyze premium revenue, claims cost, and overall portfolio profitability using SQL queries and business intelligence visualizations.

2. Dataset Generation Approach

- Two datasets were created: Policy Sales Data and Claims Data.
- A total of 1,000,000 policy records were simulated to represent customers purchasing car insurance policies in 2024.
- Policy tenure distribution followed the given assumptions: 20% (1-year), 30% (2-year), 40% (3-year), and 10% (4-year).
- Vehicle value was assumed constant at ₹100,000 for all vehicles.
- Policy premium was calculated as ₹100 per year of policy tenure.
- Policy start date was defined as 365 days after the purchase date.
- Policy end date was calculated based on policy tenure duration.

3. Claims Simulation Logic

- In 2025, 30% of vehicles purchased on the 7th, 14th, 21st, and 28th of each month were simulated to file a claim.
- These claims were assumed to occur exactly on the policy start date.
- In the period between January 1, 2026 and February 28, 2026, 10% of vehicles with a 4-year policy tenure were simulated to file an additional claim.
- Claim amounts were assumed to be 10% of the vehicle value (₹10,000).
- Each vehicle can file only one claim per year, but vehicles eligible in both periods may file a second claim.

4. Analytical Approach

- SQL queries were used to compute total premium collected during the simulation period.
- Monthly and yearly claim costs were calculated to understand claim patterns.
- Claim cost-to-premium ratios were analyzed across policy tenure categories.
- Claim ratios were also evaluated based on the month in which policies were sold.
- Future claim liabilities were estimated for policies that have not yet filed claims.
- Premium earning analysis was performed using daily premium allocation over policy tenure.

5. Dashboard Visualization

- A Power BI dashboard was designed to visualize portfolio performance metrics.
- Key performance indicators include Total Policies, Total Premium Collected, Total Claim Cost, and Loss Ratio.
- Visualizations include monthly claim trend charts, claim ratio by policy tenure, and premium versus claim comparison charts.
- The dashboard layout prioritizes readability, structured analysis, and business decision support.

6. Key Assumptions

- Vehicle value is fixed at ₹100,000 for all customers.
- Premium pricing is uniform at ₹100 per policy year.
- Policy sales are evenly distributed throughout the year.
- Claims occur strictly based on the defined defect scenarios.
- Operational costs, taxes, and administrative expenses are excluded from the analysis.

7. Conclusion

This assignment demonstrates how simulated insurance datasets can be structured, analyzed, and visualized to evaluate business performance. By combining SQL-based analysis with Power BI dashboards, the project highlights key insights into revenue generation, risk exposure, and portfolio profitability.